Chemically Responsive Active Swimmers

From hybrid particle dynamics to macroscopic transport

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1 Overview

1.1 What these notes do

A single self-propelled particle in two dimensions is allowed to sense a chemical field and adjust any of its four kinetic parameters in response. There are two distinct things it can sense and four parameters it can adjust, giving eight elementary response mechanisms. The aim here is to show that these eight are not eight separate calculations. They are eight settings of one model, and a single chain of steps carries all of them at once from individual trajectories to a macroscopic transport equation.

The chain has four links, and the notes follow them in order:

1. a stochastic process for one particle (section 3);
2. the partial differential equation obeyed by the probability density of that process (section 4);
3. angular moments of that density, closed by a systematic approximation (section 5);
4. the boundary condition consistent with the closure (section 6).

The end product is a drift–diffusion equation with two coefficients, an effective diffusivity and a chemotactic mobility, both expressed through the microscopic parameters. Reading off which of the eight mechanisms contribute to each coefficient is then a matter of inspection.

1.2 Standing assumptions

Four assumptions hold throughout and are not repeated later.

1. *The chemical field is static and externally prescribed.* It is a smooth given function $c(\boldsymbol{r})$ of position alone; the swimmers neither produce nor consume it, and it does not evolve. Feedback of the population onto the field would make the final equation nonlinear, which is outside the scope here. The kinetic coefficients therefore have no *explicit* time dependence; along a trajectory they still vary, through $\boldsymbol{r}_t$ and θ_t.
2. *Responses are weak.* Each kinetic parameter deviates only slightly from its background value, so a first-order expansion in the chemical signal suffices.
3. *Reorientation is unbiased and achiral.* The probability of turning from one heading to another depends only on the turn angle between them, so no direction is singled out in the laboratory frame; and left and right turns are equally likely, so no handedness is singled out either. Both halves are needed, and section 3.3 shows exactly what the second one buys.
4. *All stochastic integrals are Itô.* This is a modelling choice rather than bookkeeping: the same physical trajectory written in a different convention carries a compensating deterministic drift, so every coefficient below is meaningful only together with the Itô rule.

Note that assumption 2 is itself an approximation, and it is made at the very start: equation (2) replaces the true response functions by their linearizations. Everything up to section 5 is exact *for that microscopic model*, not for an arbitrary one. Two further approximations — linearity in the chemical amplitude of the resulting moment equations, and separation of time scales — enter only at the closure, and are introduced explicitly there.

2 Setup: state, signals, and response coefficients

2.1 State variables

A swimmer is described by its position and the direction it is pointing:

$$\boldsymbol{r}_t = (x_t, y_t)^{\mathsf{T}} \in \mathbb{R}^2, \qquad \theta_t \in \mathbb{S}^1 = [0, 2\pi).$$

The angle θ_t is measured from the x-axis. It is convenient to name the corresponding unit vectors,

$$\boldsymbol{e}(\theta) = \begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix}, \qquad \boldsymbol{e}_\perp(\theta) = \begin{pmatrix} -\sin\theta \\ \cos\theta \end{pmatrix},$$

so $\boldsymbol{e}$ is the heading and $\boldsymbol{e}_\perp$ is perpendicular to it. Two identities that will be used without further comment follow by direct differentiation:

$$\partial_\theta \boldsymbol{e} = \boldsymbol{e}_\perp, \qquad \partial_\theta^2 \boldsymbol{e} = -\boldsymbol{e}.$$

2.2 The chemical field and the two sensory signals

Let $c(\boldsymbol{r}) > 0$ be the chemical concentration, a fixed function of position, and let c_0 be a uniform reference value, for instance the background far from any source. A swimmer at $(\boldsymbol{r}, \theta)$ has access to two scalars built from c:

$$\begin{aligned} \delta c(\boldsymbol{r}) &= c(\boldsymbol{r}) - c_0 \qquad && \textit{concentration signal}, \\ S(\boldsymbol{r}, \theta) &= \boldsymbol{e}(\theta) \cdot \boldsymbol{\nabla} c(\boldsymbol{r}) \qquad && \textit{alignment signal}. \end{aligned} \tag{1}$$

The distinction between them drives everything that follows. The concentration signal δc is a property of the position alone: rotating the swimmer in place does not change it. The alignment signal S does depend on the heading, and changes sign when the swimmer turns around; $S > 0$ means the swimmer is currently pointing up-gradient. In the language used below, δc is *scalar* and S is *polar*.

The consequence, anticipated here and derived in section 5.6, is that a scalar signal cannot on its own single out a direction in space, whereas a polar one can. Mechanisms driven by δc therefore need a second ingredient before they can generate directed motion, while mechanisms driven by S do not.

2.3 The four kinetic parameters

The swimmer's motion is controlled by four quantities:

Symbol	Name	Role
v	swimming speed	magnitude of self-propulsion along $\boldsymbol{e}$
D_T	translational diffusivity	amplitude of positional noise
D_R	rotational diffusivity	amplitude of continuous heading noise
λ	tumble rate	frequency of discrete reorientation events

Each is written as a background value plus a linear response to the two signals. Using X as a placeholder for any of the four,

$$X(\boldsymbol{r}, \theta) = X_0 + X^c\, \delta c(\boldsymbol{r}) + X^S\, S(\boldsymbol{r}, \theta), \qquad X \in \{D_T, D_R, v, \lambda\}. \tag{2}$$

This introduces three new symbols per parameter:

- $X_0 \in \{v_0, D_{T0}, D_{R0}, \lambda_0\}$, the *background value*, attained when $c \equiv c_0$;
- X^c, the *concentration sensitivity*, i.e. $\partial X/\partial \delta c$ evaluated at the background;
- X^S, the *alignment sensitivity*, i.e. $\partial X/\partial S$ evaluated at the background.

There are eight sensitivities in total, four of each kind. Superscripts c and S are labels, never exponents.

2.4 Admissibility of the linearization

All four parameters must stay non-negative: a negative diffusivity or tumble rate is meaningless. Since $|S| = |\boldsymbol{e} \cdot \boldsymbol{\nabla} c| \le |\boldsymbol{\nabla} c|$, a sufficient condition over a prescribed range of signals is

$$|X^c| \, \|\delta c\|_\infty + |X^S| \, \|\boldsymbol{\nabla} c\|_\infty \le X_0, \tag{3}$$

where $\|\cdot\|_\infty$ denotes the supremum over the domain. This is a range-of-validity statement about equation (2), not a physical law, and it should not be over-read: it constrains each sensitivity relative to the signal range on which that sensitivity was specified, which is not the same as bounding the macroscopic drift.

The gradient-driven terms do inherit a bound, as noted in section 5.5, but the concentration-driven ones do not. The reason is a mismatch of norms. Equation (3) controls $|D_T^c| \, \|\delta c\|_\infty$, whereas the drift it generates involves $|D_T^c| \, \|\boldsymbol{\nabla} c\|_\infty$, and the ratio $\|\boldsymbol{\nabla} c\|_\infty / \|\delta c\|_\infty$ can be made arbitrarily large by a field that is steep but shallow in amplitude.

3 The microscopic process

The swimmer alternates between smooth stochastic motion and instantaneous reorientation events. These are described separately and then combined.

3.1 Motion between tumbles

Away from tumbles the swimmer does two things at once: it drives itself forward along its current heading, and it is buffeted by noise in both position and orientation. Let $\boldsymbol{W}_t$ be a two-dimensional Wiener process driving positional noise and W_t^R an independent one-dimensional Wiener process driving heading noise. Then

$$\begin{aligned} \mathrm{d}\boldsymbol{r}_t &= \underbrace{v \, \boldsymbol{e}(\theta_t) \, \mathrm{d}t}_{\text{self-propulsion}} + \underbrace{\sqrt{2D_T} \, \mathrm{d}\boldsymbol{W}_t}_{\text{positional noise}}, \\ \mathrm{d}\theta_t &= \underbrace{\sqrt{2D_R} \, \mathrm{d}W_t^R}_{\text{heading noise}}, \end{aligned} \tag{4}$$

with v, D_T, D_R evaluated at $(\boldsymbol{r}_t, \theta_t)$ through equation (2). Note what is *not* here: there is no deterministic torque, so between tumbles the heading changes only diffusively, and no external force, so the only systematic motion is the swimmer's own propulsion.

The factors of $\sqrt{2}$ fix the normalization of the diffusivities. Over a short interval, with the drift switched off, equation (4) gives $\langle |\Delta \boldsymbol{r}|^2 \rangle = 4 D_T \, \Delta t$ in two dimensions (that is $2 D_T \Delta t$ per Cartesian component) and $\langle (\Delta\theta)^2 \rangle = 2 D_R \, \Delta t$. So D_T carries units of length^2/time and D_R of 1/time; D_R is a rate, not a diffusivity in space, which is why it can be compared directly with λ later.

3.2 Tumbles

A tumble is instantaneous, so it cannot be added to equation (4) as another Langevin term: it is a jump, not a drift or a diffusion. Three objects are needed to specify it, and it is worth being explicit about what each one is, because they are easy to conflate.

N: which tumbles happen, and where they land. A single realization of the swimmer's motion contains a discrete list of tumble events. Label them $(t_1, \theta_1), (t_2, \theta_2), \ldots$, meaning "at time t_k the swimmer tumbled, and the heading it drew was θ_k". That list is exactly the content of N: it is a *marked counting measure* on $[0, \infty) \times \mathbb{S}^1$, a random scatter of points, one per tumble, whose first coordinate is the time and whose second coordinate — the *mark* — is the new heading. For any region A of that space, $N(A)$ is the integer number of points falling in A, so N literally counts events. In differential notation, $N(\mathrm{d}t, \mathrm{d}\theta')$ equals 1 if the list contains a point in the cell $[t, t + \mathrm{d}t) \times [\theta', \theta' + \mathrm{d}\theta')$ and 0 otherwise, and over a short enough interval at most one cell is occupied.

This is what makes the jump term in equation (6) work. The integral $\int_0^{2\pi} \mathrm{wrap}(\theta' - \theta_{t^-})\, N(\mathrm{d}t, \mathrm{d}\theta')$ vanishes on any interval containing no tumble, since N has no points there; at a tumble with mark θ_k it contributes the single increment $\mathrm{wrap}(\theta_k - \theta_{t_k^-})$, which resets the heading to θ_k.

λ and T: when, and where to. N is not yet specified as a random object; what remains is to say how likely its points are. This is done by its *compensator*, the expected number of points per unit time and per unit angle, conditioned on everything that has happened so far:

$$\mathbb{E}\big[N(\mathrm{d}t, \mathrm{d}\theta') \mid \text{past}\big] = \underbrace{\lambda(\boldsymbol{r}_{t^-}, \theta_{t^-})}_{\text{when}}\ \underbrace{T(\theta' \mid \theta_{t^-})}_{\text{where to}}\, \mathrm{d}\theta'\, \mathrm{d}t, \qquad \int_0^{2\pi} T(\theta' \mid \theta)\, \mathrm{d}\theta' = 1. \tag{5}$$

The factorization is the whole point, and the two factors answer different questions and carry different units.

The *tumble rate* λ, with units of 1/time, answers *when*. Integrating equation (5) over all marks and using the normalization of T leaves $\lambda\, \mathrm{d}t$: the probability of tumbling at all during $\mathrm{d}t$, irrespective of outcome. Equivalently, the probability of running without interruption from 0 to t is $\exp[-\int_0^t \lambda(\boldsymbol{r}_s, \theta_s)\, \mathrm{d}s]$, so for constant λ_0 run durations are exponential with mean $1/\lambda_0$, and a swimmer that raises λ shortens its runs.

The *tumble kernel* $T(\theta' \mid \theta)$, with units of 1/angle, answers *where to*. Given that a tumble has occurred and that the swimmer was heading along θ, it is the probability density of the new heading θ'. Its normalization states that a tumble always yields some heading: T redistributes headings but neither creates nor destroys swimmers. Complete randomization is $T = 1/2\pi$; a swimmer that turns only slightly has T sharply peaked at $\theta' = \theta$.

Because λ and T are separate, a swimmer can modulate *how often* it turns without changing *how far* it turns, and vice versa. Case 8 of table 1 modulates the first; the persistence parameter of section 3.3 describes the second.

The complete process. Combining equation (4) with the jump term,

$$\begin{aligned} \mathrm{d}\boldsymbol{r}_t &= v\, \boldsymbol{e}(\theta_{t^-})\, \mathrm{d}t + \sqrt{2D_T}\, \mathrm{d}\boldsymbol{W}_t, \\ \mathrm{d}\theta_t &= \sqrt{2D_R}\, \mathrm{d}W_t^R + \int_0^{2\pi} \mathrm{wrap}(\theta' - \theta_{t^-})\, N(\mathrm{d}t, \mathrm{d}\theta') \pmod{2\pi}, \end{aligned} \tag{6}$$

which is the microscopic model in full. Two notational points remain.

The subscript t^- is the left limit, the state immediately *before* any jump at time t. Every coefficient in equations (5) and (6) is evaluated there. This makes them *predictable*: their values are fixed by the past and not by the outcome of the jump itself. Physically it says the swimmer decides whether to turn on the basis of what it was sensing while running, which is the only causal option. Mathematically it is also what makes equation (5) define a genuine process, and, as section 4.2 shows, what makes the gain and loss terms of the kinetic equation cancel so that probability is conserved.

The function $\mathrm{wrap}(\alpha) \in (-\pi, \pi]$ returns the representative of an angular increment on the circle, needed because θ lives on $\mathbb{S}^1$ and the difference of two angles is defined only modulo 2π. It is pure bookkeeping: the equivalent statement "at a jump with mark θ', set $\theta_t = \theta'$" says the same thing without the device.

3.3 Directional persistence of the tumble kernel

Assumption 3 of section 2 has two halves, and they do different work.

Rotation invariance says T can depend only on the turn angle,

$$T(\theta' \mid \theta) = \widehat{T}(\theta' - \theta),$$

so no laboratory direction is preferred. *Achirality* says $\widehat{T}$ is an even function,

$$\widehat{T}(\psi) = \widehat{T}(-\psi),$$

so left and right turns of the same size are equally likely.

Only one number from $\widehat{T}$ survives at the order worked to here, and it is the mean cosine of the turn,

$$a_1 = \int_0^{2\pi} \cos\psi \, \widehat{T}(\psi) \, \mathrm{d}\psi \;=\; \langle \cos(\text{turn angle}) \rangle. \tag{7}$$

Two limits fix its meaning. Complete randomization, $\widehat{T} = 1/2\pi$, makes every new heading equally likely, the swimmer forgets where it was pointing, and $a_1 = 0$. Small typical turns leave the new heading close to the old, and a_1 approaches 1. Values near 0.3 are commonly quoted for *E. coli*.

The way a_1 enters is through the first angular moment of the kernel. Writing $\theta = \theta' + \psi$ and using $\boldsymbol{e}(\theta' + \psi) = \cos\psi \, \boldsymbol{e}(\theta') + \sin\psi \, \boldsymbol{e}_\perp(\theta')$,

$$\int_0^{2\pi} \boldsymbol{e}(\theta) \, T(\theta \mid \theta') \, \mathrm{d}\theta = a_1 \, \boldsymbol{e}(\theta') + b_1 \, \boldsymbol{e}_\perp(\theta'), \qquad b_1 = \int_0^{2\pi} \sin\psi \, \widehat{T}(\psi) \, \mathrm{d}\psi.$$

This is where achirality is needed, and rotation invariance alone would not suffice: a chiral kernel has $b_1 \neq 0$ and leaves a component of the averaged post-tumble heading *perpendicular* to the pre-tumble one, which would propagate into the polarity equation and into κ as transverse terms describing systematic circling. Evenness of $\widehat{T}$ kills the $\sin\psi$ integrand, so $b_1 = 0$ and

$$\int_0^{2\pi} \boldsymbol{e}(\theta) \, T(\theta \mid \theta') \, \mathrm{d}\theta = a_1 \, \boldsymbol{e}(\theta'). \tag{8}$$

In words: averaging the post-tumble heading over the kernel returns the pre-tumble heading, shrunk by the factor a_1 and not rotated.

3.4 The eight canonical cases

The response model equation (2) contains eight sensitivities. Switching on exactly one at a time, and holding the other seven at zero, isolates eight elementary mechanisms. They are listed in table 1: cases 1–4 are driven by the scalar signal, cases 5–8 by the polar one.

Case	Sensitivity	Common name	What the swimmer does
1	D_T^c	concentration-dependent diffusion	Positional noise amplitude varies with local concentration.
2	D_R^c	concentration-dependent rotational noise	Heading noise, hence orientational memory, varies with concentration.
3	v^c	orthokinesis	Run speed varies with local concentration.
4	λ^c	concentration-dependent tumbling	Run duration varies with concentration, ignoring heading.
5	D_T^S	alignment-gated positional noise	Positional noise is scaled by heading, but stays isotropic.
6	D_R^S	alignment-gated rotational noise	Headings aligned with the gradient are stabilized or destabilized.
7	v^S	alignment-gated propulsion	Speed differs between up- and down-gradient headings.
8	λ^S	local run-and-tumble surrogate	Runs are extended or shortened according to the heading.

Table 1: The eight canonical cases. Terminology for cases 5–7 varies between fields, so the defining sensitivity is less ambiguous than a biological label.

Case 8 deserves a word of caution. The rule $\lambda = \lambda(S)$ reproduces the local effect of suppressing tumbles while heading up-gradient, and in that narrow sense it captures bacterial chemotaxis. It is nonetheless a surrogate: real *E. coli* compares concentrations over time using an adaptive response with memory, rather than reading an instantaneous gradient projection. The difference is quantitative as well as conceptual, and one should not import results derived for memory-based models into this one without checking them.

4 The kinetic equation

4.1 From one trajectory to an ensemble

Together, equations (5) and (6) describe what one swimmer does. They are awkward to work with directly: each realization is a different random path, whereas the quantities we ultimately care about — how many swimmers sit where, and which way they point — are properties of the ensemble rather than of any single path. So we change the object of study.

Let $P(\boldsymbol{r}, \theta, t)$ be the joint probability density of position and heading, normalized so that $P\,\mathrm{d}^2r\,\mathrm{d}\theta$ is the probability of finding the swimmer in the phase-space element $\mathrm{d}^2r\,\mathrm{d}\theta$ at time t. For an ensemble of non-interacting swimmers, P is equally well read as their phase-space number density, and the two readings differ only by a constant. The aim of this section is the partial differential equation that P obeys.

Three distinct mechanisms move probability around, and it is worth separating them before combining them:

- *self-propulsion*, which transports probability through space at velocity $v\boldsymbol{e}(\theta)$ without changing θ;
- *diffusion*, which spreads probability in $\boldsymbol{r}$ and in θ continuously;
- *tumbling*, which teleports probability from one heading to another at fixed $\boldsymbol{r}$.

The first two are continuous and produce derivatives of P; the third is discontinuous and produces an integral over headings. Handling the third is what makes this a *master* equation as well as a

Fokker–Planck equation, and it is treated first because it is the least familiar.

4.2 The tumble terms: a master equation in heading space

Freeze the position and switch off all diffusion, so that the heading is the only variable and it changes only by tumbling. Over a short interval Δt, a swimmer currently on heading θ either tumbles or does not. From equation (5) the probability of tumbling at all is $\lambda(\theta)\,\Delta t + o(\Delta t)$, and given that it does, the probability of landing in $\mathrm{d}\theta$ about θ having started from θ' is $T(\theta \mid \theta')\,\mathrm{d}\theta$.

Book-keeping over the two ways of being on heading θ at time $t + \Delta t$ — either being there already and not tumbling, or tumbling in from somewhere else — gives

$$P(\theta, t+\Delta t) = \underbrace{\big[1 - \lambda(\theta)\Delta t\big]\,P(\theta,t)}_{\text{stayed}} + \underbrace{\Delta t \int_0^{2\pi} \lambda(\theta')\,T(\theta \mid \theta')\,P(\theta',t)\,\mathrm{d}\theta'}_{\text{arrived}} + o(\Delta t).$$

Subtracting $P(\theta, t)$, dividing by Δt and letting $\Delta t \to 0$ gives the master equation

$$\partial_t P(\theta,t) = \int_0^{2\pi} \lambda(\theta')\,T(\theta \mid \theta')\,P(\theta',t)\,\mathrm{d}\theta' - \lambda(\theta)\,P(\theta,t). \tag{9}$$

The two terms have names. The integral is the *gain*: it sums, over every possible source heading θ', the rate at which swimmers leave θ' and the fraction of those that land on θ. The second term is the *loss*: swimmers currently on θ leaving it at rate $\lambda(\theta)$.

Two features of equation (9) repay attention, because both are easy to get wrong and both matter later.

First, *the rate inside the gain integral is evaluated at the source heading θ', not at θ.* This is the mathematical expression of the pre-jump evaluation discussed in section 3: what governs how many swimmers arrive on θ is how fast swimmers were leaving θ', which depends on the signal those swimmers were sensing before they turned. Writing $\lambda(\theta)$ inside the integral would say instead that the arrival rate depends on where the swimmer is going, which is not causal.

Second, *equation* (9) *conserves probability, and the source evaluation is exactly what makes it do so.* Integrating over θ and exchanging the order of integration in the gain term,

$$\int_0^{2\pi}\!\!\int_0^{2\pi} \lambda(\theta')T(\theta \mid \theta')P(\theta')\,\mathrm{d}\theta'\,\mathrm{d}\theta = \int_0^{2\pi} \lambda(\theta')P(\theta') \underbrace{\left[\int_0^{2\pi} T(\theta \mid \theta')\,\mathrm{d}\theta\right]}_{=1} \mathrm{d}\theta' = \int_0^{2\pi} \lambda\,P\,\mathrm{d}\theta,$$

which cancels the loss term identically. The normalization of T in equation (5) is what collapses the inner bracket, and it is the statement that a tumble always produces *some* heading. Had λ been evaluated at θ inside the integral it would not have factored out of the inner integral, the cancellation would have failed, and the equation would silently create or destroy swimmers.

4.3 The continuous terms, and where the diffusivities sit

Now switch tumbling off and restore the continuous motion of equation (6). Probability is conserved locally, so P obeys a continuity equation in phase space,

$$\partial_t P = -\boldsymbol{\nabla}\cdot\boldsymbol{J} - \partial_\theta j_\theta,$$

where $\boldsymbol{J}$ is the spatial probability current and j_θ the angular one. Self-propulsion contributes an advective current $v\,\boldsymbol{e}P$. The diffusive contributions are where care is needed, and the answer depends on the stochastic convention.

For the Itô convention the currents are

$$\boldsymbol{J} = v\,\boldsymbol{e}P - \boldsymbol{\nabla}\big(D_T P\big), \qquad j_\theta = -\partial_\theta\big(D_R P\big),$$

with the diffusivities *inside* the derivatives. This is not a matter of taste. If D_T varies in space, then $\boldsymbol{\nabla}(D_T P)$ and $D_T\boldsymbol{\nabla}P$ differ by $P\boldsymbol{\nabla}D_T$, a genuine current with observable consequences: at zero flux the first gives $P \propto 1/D_T$, whereas the second gives P uniform. The next subsection shows why Itô produces the first form.

4.4 The systematic route: generator and adjoint

The reasoning above is transparent but assembled by hand. There is a systematic procedure that produces all the terms at once and settles the placement of the diffusivities without guesswork. It runs in two steps.

Step one: the backward generator. For a smooth test function $f(\boldsymbol{r},\theta)$, periodic in θ, define

$$\mathcal{L}f(\boldsymbol{r},\theta) = \lim_{\Delta t\downarrow 0}\frac{1}{\Delta t}\Big(\mathbb{E}\big[f(\boldsymbol{r}_{t+\Delta t},\theta_{t+\Delta t}) \mid \boldsymbol{r}_t = \boldsymbol{r},\ \theta_t = \theta\big] - f(\boldsymbol{r},\theta)\Big),$$

the expected rate of change of f along a trajectory started at $(\boldsymbol{r},\theta)$. Applying the Itô–Lévy formula to equation (6) — the drift and Wiener terms by Itô's lemma, the jump term by averaging f over the tumble kernel — gives

$$\mathcal{L}f = v\,\boldsymbol{e}\cdot\boldsymbol{\nabla}f + D_T\boldsymbol{\nabla}^2 f + D_R\partial_\theta^2 f + \lambda\int_0^{2\pi}\big[f(\boldsymbol{r},\theta') - f(\boldsymbol{r},\theta)\big]T(\theta' \mid \theta)\,\mathrm{d}\theta'. \tag{10}$$

Every coefficient in equation (10) sits *outside* the derivatives acting on f. That placement is the structural signature of the Itô convention: Itô's lemma evaluates the noise amplitude at the start of each increment, so no derivative of D_T or D_R is generated.

Step two: the adjoint. The generator describes f; we want an equation for P. The bridge is that the two descriptions must agree on expectation values. Abbreviating the phase-space integral as

$$\int(\,\cdot\,)\,\mathrm{d}\Gamma \;\equiv\; \int_{\mathbb{R}^2}\mathrm{d}^2 r\int_0^{2\pi}\mathrm{d}\theta\;(\,\cdot\,),$$

that agreement reads

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathbb{E}[f] = \int f\,\partial_t P\,\mathrm{d}\Gamma = \int(\mathcal{L}f)\,P\,\mathrm{d}\Gamma.$$

Moving every derivative off f and onto P by integration by parts, and discarding the boundary terms at infinity, defines the adjoint operator $\mathcal{L}^\dagger$ through $\int(\mathcal{L}f)\,P\,\mathrm{d}\Gamma = \int f\,(\mathcal{L}^\dagger P)\,\mathrm{d}\Gamma$. Term by term,

$$\int P\,\big(v\,\boldsymbol{e}\cdot\boldsymbol{\nabla}f\big)\,\mathrm{d}\Gamma = -\int f\,\boldsymbol{\nabla}\cdot\big(v\,\boldsymbol{e}P\big)\,\mathrm{d}\Gamma,$$

$$\int P\,\big(D_T\boldsymbol{\nabla}^2 f\big)\,\mathrm{d}\Gamma = +\int f\,\boldsymbol{\nabla}^2\big(D_T P\big)\,\mathrm{d}\Gamma,$$

$$\int P\,\big(D_R\partial_\theta^2 f\big)\,\mathrm{d}\Gamma = +\int f\,\partial_\theta^2\big(D_R P\big)\,\mathrm{d}\Gamma,$$

$$\int P\,\lambda\int_0^{2\pi} f(\theta')\,T(\theta' \mid \theta)\,\mathrm{d}\theta'\,\mathrm{d}\Gamma = \int f\int_0^{2\pi}\lambda(\theta')\,T(\theta \mid \theta')\,P(\theta')\,\mathrm{d}\theta'\,\mathrm{d}\Gamma.$$

The first line uses one integration by parts, which supplies the minus sign. The second and third use two, so the signs cancel; the boundary terms in θ vanish by periodicity. The fourth is not

an integration by parts at all. It merely exchanges the names of the two dummy angles, turning the generator's difference operator into the gain and loss pair of equation (9).

The second and third lines are the point of the exercise. Because D_T and D_R began *outside* the derivatives acting on f, the two integrations by parts deposit them *inside* the derivatives acting on P. This is where the $\boldsymbol{\nabla}(D_T P)$ current asserted above comes from, and it is why cases 1 and 6 — the two mechanisms that modulate a noise amplitude — carry the coefficients they do in the final answer.

4.5 The full kinetic equation

Collecting, and writing $\partial_t P = \mathcal{L}^\dagger P$,

$$\partial_t P = \underbrace{-\boldsymbol{\nabla}\cdot(v\,\boldsymbol{e}P)}_{\text{self-propulsion}} + \underbrace{\boldsymbol{\nabla}^2(D_T P) + \partial_\theta^2(D_R P)}_{\text{diffusion}} + \underbrace{\int_0^{2\pi} \lambda(\boldsymbol{r},\theta')\,T(\theta\mid\theta')\,P(\boldsymbol{r},\theta',t)\,\mathrm{d}\theta' - \lambda(\boldsymbol{r},\theta)\,P}_{\text{tumbling}}. \tag{11}$$

All coefficients are evaluated at $(\boldsymbol{r},\theta)$ except inside the gain integral, where they are evaluated at the source heading θ'.

This equation is exact for the microscopic model of equations (5) and (6): nothing in the derivation expanded in the chemical signal, and it holds for arbitrary response functions v, D_T, D_R, λ. Within these notes those functions are the linearizations equation (2), which is the one approximation already in force. All eight cases of table 1 are contained here, and everything in the remaining sections is an approximation to this single equation.

5 Coarse-graining: moments and closure

5.1 Angular moments

The kinetic equation lives on a three-dimensional space. Integrating out the heading leaves fields on physical space alone. The first three angular moments are

$$\rho(\boldsymbol{r},t) = \int_0^{2\pi} P\,\mathrm{d}\theta, \qquad \boldsymbol{p}(\boldsymbol{r},t) = \int_0^{2\pi} \boldsymbol{e}\,P\,\mathrm{d}\theta, \qquad \boldsymbol{Q}(\boldsymbol{r},t) = \int_0^{2\pi} \Big(\boldsymbol{e}\boldsymbol{e}^{\mathsf{T}} - \tfrac{1}{2}\boldsymbol{I}\Big)P\,\mathrm{d}\theta,$$

where $\boldsymbol{I}$ is the 2×2 identity. Their meanings are: ρ is the particle density; $\boldsymbol{p}$ is the *polarity*, the mean heading weighted by density, which is nonzero only if more swimmers point one way than another; and $\boldsymbol{Q}$ is the traceless *nematic tensor*, which is nonzero if swimmers align along an axis without preferring either direction along it.

Two identities used repeatedly below follow from $\boldsymbol{e}\boldsymbol{e}^{\mathsf{T}} + \boldsymbol{e}_\perp\boldsymbol{e}_\perp^{\mathsf{T}} = \boldsymbol{I}$:

$$\int_0^{2\pi} \boldsymbol{e}\boldsymbol{e}^{\mathsf{T}} P\,\mathrm{d}\theta = \boldsymbol{Q} + \tfrac{1}{2}\rho\boldsymbol{I}, \qquad \int_0^{2\pi} \boldsymbol{e}_\perp\boldsymbol{e}_\perp^{\mathsf{T}} P\,\mathrm{d}\theta = \tfrac{1}{2}\rho\boldsymbol{I} - \boldsymbol{Q}.$$

5.2 The ordering scheme

Moment equations never close on their own: the equation for ρ involves $\boldsymbol{p}$, that for $\boldsymbol{p}$ involves $\boldsymbol{Q}$, and so on. Truncating requires knowing which terms are small, so two bookkeeping parameters are introduced.

Chemical amplitude ε. Write $c = c_0 + \varepsilon c_1$, so that δc, $\boldsymbol{\nabla}c$ and S are all $\mathcal{O}(\varepsilon)$. Expanding about a homogeneous reference state then gives $\rho = \rho_0 + \mathcal{O}(\varepsilon)$, and with it $\boldsymbol{\nabla}\rho = \mathcal{O}(\varepsilon)$. The point

is that with no chemistry there is nothing to structure the population at all: every departure from uniformity is driven by c, and so carries at least one power of ε.

Scale separation δ. Let L be the length over which c and ρ vary appreciably and let

$$\ell_p = \frac{v_0}{\gamma_0}$$

be the persistence length, the distance a swimmer covers before forgetting its heading, with the relaxation rate γ_0 defined in equation (13) below. Set $\delta = \ell_p/L \ll 1$.

The two parameters combine to grade the moment hierarchy. Anticipating equation (15), the slaved polarity is of size $\boldsymbol{p} \sim (v_0/\gamma_0)|\boldsymbol{\nabla}\rho| = \ell_p|\boldsymbol{\nabla}\rho|$, so

$$\frac{\boldsymbol{p}}{\rho} = \mathcal{O}(\varepsilon\delta), \qquad \frac{\boldsymbol{Q}}{\rho} = \mathcal{O}(\varepsilon\delta^2),$$

the second following because $\boldsymbol{Q}$ is sourced by gradients of $\boldsymbol{p}$ and so carries one further power of δ. Each successive moment is suppressed by one more factor of δ; this, not the chemical amplitude alone, is what justifies truncating at $\boldsymbol{p}$.

The closure retains terms linear in ε at leading order in δ. Three products therefore drop out:

$$\delta c\,\boldsymbol{p}, \qquad \boldsymbol{p}\cdot\boldsymbol{\nabla}c = \mathcal{O}(\varepsilon^2\delta), \qquad \boldsymbol{Q}\boldsymbol{\nabla}c = \mathcal{O}(\varepsilon^2\delta^2),$$

the first being smaller than the retained $v_0\boldsymbol{p}$ by one power of ε. This ordering fixes the scope of the macroscopic equation alone. Nothing before section 5 depends on it.

5.3 The exact density equation

Integrating equation (11) over θ gives a conservation law, since the tumble terms cancel and the ∂_θ^2 term vanishes by periodicity:

$$\partial_t\rho = -\boldsymbol{\nabla}\cdot\boldsymbol{J}_\rho.$$

The mass flux $\boldsymbol{J}_\rho$ follows from the moment identities above and is still exact:

$$\boldsymbol{J}_\rho = (v_0 + v^c\delta c)\,\boldsymbol{p} + v^S\Big(\boldsymbol{Q} + \tfrac{1}{2}\rho\boldsymbol{I}\Big)\boldsymbol{\nabla}c - \boldsymbol{\nabla}\Big[(D_{T0} + D_T^c\delta c)\,\rho + D_T^S\,\boldsymbol{p}\cdot\boldsymbol{\nabla}c\Big]. \tag{12}$$

Two structural observations can be made before any approximation.

First, D_{R0} and λ_0 do not appear. Reorientation shuffles headings without transporting mass, so it cannot enter the flux directly; it acts on ρ only indirectly, by controlling how quickly $\boldsymbol{p}$ settles.

Second, two mechanisms can drive a current from a perfectly isotropic heading distribution. Setting $\boldsymbol{p} = \boldsymbol{Q} = \boldsymbol{0}$ and taking ρ uniform, equation (12) leaves

$$\boldsymbol{J}_\rho\big|_{\boldsymbol{p}=\boldsymbol{Q}=\boldsymbol{0}} = \Big(\tfrac{1}{2}v^S - D_T^c\Big)\rho\boldsymbol{\nabla}c,$$

so cases 7 and 1 both act *directly*. For case 7 the reading is immediate: if up-gradient swimmers move faster than down-gradient ones there is net transport with no excess of swimmers pointing either way. For case 1 the current comes from the Itô divergence form $-\boldsymbol{\nabla}(D_T\rho)$, which does not vanish for uniform ρ when D_T varies in space. Cases 3, 6 and 8 have no such term and must work through $\boldsymbol{p}$. The distinction is visible in the final answer: in equation (18) the contributions $v^S/2$ and $-D_T^c$ stand alone, whereas those of v^c, D_R^S and λ^S all carry the prefactor $v_0/2\gamma_0$ inherited from the slaved polarity.

5.4 The polarity equation

Multiplying equation (11) by $\boldsymbol{e}(\theta)$ and integrating over $\theta \in [0, 2\pi)$ gives the evolution of $\boldsymbol{p}$. Two contributions set the rate at which polarity decays. Rotational diffusion contributes $-D_{R0}\boldsymbol{p}$, using $\partial_\theta^2 \boldsymbol{e} = -\boldsymbol{e}$ after two integrations by parts. Tumbling contributes $-\lambda_0 \boldsymbol{p}$ from the loss term and $+a_1\lambda_0\boldsymbol{p}$ from the gain term, by equation (8). Together they define the *polarity relaxation rate*

$$\gamma_0 = D_{R0} + \lambda_0\,(1 - a_1). \tag{13}$$

The factor $1 - a_1$ is worth pausing on: a tumble rate is not the same thing as an orientation-decorrelation rate. Frequent but gentle turns (λ_0 large, a_1 close to 1) decorrelate the heading slowly.

Carrying the same calculation through for the source terms, to first order in ε and leading order in δ,

$$\partial_t \boldsymbol{p} + \gamma_0 \boldsymbol{p} = -\frac{v_0}{2}\boldsymbol{\nabla}\rho - \frac{v^c + D_R^S + (1 - a_1)\lambda^S}{2}\,\rho\boldsymbol{\nabla}c. \tag{14}$$

Two features of equation (14) are easy to get wrong.

The coefficient v^c appears exactly *once*. It arises from expanding $-\frac{1}{2}\boldsymbol{\nabla}[(v_0 + v^c\delta c)\rho] = -\frac{1}{2}v_0\boldsymbol{\nabla}\rho - \frac{1}{2}v^c\rho\boldsymbol{\nabla}c - \frac{1}{2}v^c\delta c\boldsymbol{\nabla}\rho$, whose last term is $\mathcal{O}(\varepsilon^2)$ and is discarded. Writing both the gradient form and a separate $v^c\rho\boldsymbol{\nabla}c$ term double-counts it.

The persistence factor $1 - a_1$ multiplies λ^S in the source, in addition to appearing inside γ_0. It does not multiply D_R^S, whose contribution comes from continuous rotation rather than from tumbles. The consequence is that the two factors do not cancel in the ratio κ/D_{eff} below.

5.5 Adiabatic elimination and the closed equation

On time scales long compared with γ_0^{-1}, the polarity has relaxed to whatever the local density and gradient demand. Setting $\partial_t \boldsymbol{p} \simeq 0$ in equation (14) and $\boldsymbol{Q} \simeq 0$ by section 5.2 gives the *slaved polarity*

$$\boldsymbol{p} \simeq -\frac{v_0}{2\gamma_0}\boldsymbol{\nabla}\rho - \frac{v^c + D_R^S + (1 - a_1)\lambda^S}{2\gamma_0}\,\rho\boldsymbol{\nabla}c. \tag{15}$$

Its two terms have distinct origins. The first is a diffusive backflow: self-propelled particles stream out of crowded regions, so a density gradient alone produces net polarity. The second is genuine chemical alignment, in which a heading-dependent turning rate or a concentration-dependent speed biases the distribution of headings toward or away from the gradient.

Substituting equation (15) into equation (12) and keeping terms of order ε closes the hierarchy. The result is a drift–diffusion equation,

$$\partial_t \rho = \boldsymbol{\nabla}\cdot\big(D_{\text{eff}}\boldsymbol{\nabla}\rho - \kappa\,\rho\boldsymbol{\nabla}c\big), \tag{16}$$

with an *effective diffusivity* and a *chemotactic mobility*

$$D_{\text{eff}} = D_{T0} + \frac{v_0^2}{2\gamma_0}, \tag{17}$$

$$\kappa = \frac{v^S}{2} - D_T^c - \frac{v_0\big[v^c + D_R^S + (1 - a_1)\lambda^S\big]}{2\gamma_0}. \tag{18}$$

The two contributions to D_{eff} are ordinary translational noise and the random walk generated by persistent swimming with random reorientation. The sign convention follows from writing the flux as $\boldsymbol{J}_\rho = -D_{\text{eff}}\boldsymbol{\nabla}\rho + \kappa\rho\boldsymbol{\nabla}c$: a positive κ means drift up an attractant gradient.

One consequence of equation (3) is worth recording, in its correct and limited form. Within the prescribed signal range, $|v^S|\,\|\boldsymbol{\nabla}c\|_\infty \le v_0$, so the case-7 drift obeys $|\frac{1}{2}v^S\boldsymbol{\nabla}c| \le v_0/2$: a swimmer cannot drift faster than it swims. This is a restriction on which sensitivities are admissible

for a *given* field, not a nonlinear saturation mechanism. It does not make equation (16) a flux-limited Keller–Segel model, and it says nothing outside the range over which equation (2) was assumed to hold.

5.6 Mapping the eight cases

Reading equation (18) case by case gives table 2. Five of the eight mechanisms contribute at this order.

Case	Sensitivity	Contribution to κ	Up-gradient if	Reason
1	D_T^c	$-D_T^c$	$D_T^c < 0$	Accumulation where Itô positional noise is weaker.
2	D_R^c	0	—	A scalar signal cannot select a heading.
3	v^c	$-v_0 v^c/(2\gamma_0)$	$v^c < 0$	Slower motion in high c raises residence time.
4	λ^c	0	—	A scalar signal cannot select a heading.
5	D_T^S	0	—	Reaches ρ only via $\boldsymbol{p}\cdot\boldsymbol{\nabla}c = \mathcal{O}(\varepsilon^2\delta)$.
6	D_R^S	$-v_0 D_R^S/(2\gamma_0)$	$D_R^S < 0$	Up-gradient headings lose orientation more slowly.
7	v^S	$v^S/2$	$v^S > 0$	Faster up-gradient headings create a direct current.
8	λ^S	$-v_0(1-a_1)\lambda^S/(2\gamma_0)$	$\lambda^S < 0$	Up-gradient runs tumble less often.

Table 2: Contribution of each isolated mechanism to the chemotactic mobility κ of equation (18).

The zeros in rows 2 and 4 are the promised consequence of the scalar/polar distinction of equation (1). Modulating D_R or λ with δc changes how fast headings randomize but not which heading is favoured, so no direction is singled out. Row 5 is zero for a different reason: D_T^S reaches the density only through the combination $\boldsymbol{p}\cdot\boldsymbol{\nabla}c$, and since $\boldsymbol{p}/\rho = \mathcal{O}(\varepsilon\delta)$ the product is second order in the chemical amplitude. It is a genuine mechanism, just not a linear one.

6 Boundary conditions for a confined population

6.1 The macroscopic condition

Once the closure of section 5.5 is in force, the population is described by the single field ρ. Recall from section 5 that ρ is the marginal of P over headings, so it is a probability density per unit area, normalized to unity over the domain; for an ensemble of non-interacting swimmers it is equally a number density, the two readings differing by a constant factor that no boundary condition can see. Let $\Omega \subset \mathbb{R}^2$ be the accessible region, with boundary $\partial\Omega$ and outward unit normal $\boldsymbol{n}$, and write $\partial_n = \boldsymbol{n}\cdot\boldsymbol{\nabla}$ for the outward normal derivative.

At this level the flux is the constitutive one delivered by the closure,

$$\boldsymbol{J}_\rho = -D_{\text{eff}}\boldsymbol{\nabla}\rho + \kappa\,\rho\boldsymbol{\nabla}c, \tag{19}$$

and since equation (16) is second order in space, exactly one scalar condition is needed at each point of $\partial\Omega$.

Impermeable wall. If no swimmers cross the boundary, the normal component of equation (19) vanishes there. Written out, $\boldsymbol{n}\cdot\boldsymbol{J}_\rho = 0$ becomes

$$\partial_n\rho = \frac{\kappa}{D_{\text{eff}}}\,\rho\,\partial_n c \qquad \text{on } \partial\Omega. \tag{20}$$

This is a Robin-type condition: it fixes a combination of ρ and its normal derivative rather than either one alone, with a coefficient $(\kappa/D_{\text{eff}})\,\partial_n c$ that varies along the boundary with the local normal gradient of the attractant. Read physically, it is a local balance. Any chemotactic drift pushing swimmers into the wall must be exactly offset by a density gradient diffusing them back out, because neither can be carried away. Total number is then conserved by construction, since the divergence theorem gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_\Omega \rho\,\mathrm{d}^2 r = -\oint_{\partial\Omega} \boldsymbol{n}\cdot\boldsymbol{J}_\rho\,\mathrm{d}s = 0.$$

Note that the condition is *not* $\partial_n\rho = 0$ in general. Imposing plain Neumann on ρ where $\partial_n c \neq 0$ would force a nonzero flux through the wall and leak swimmers out of the domain.

6.2 Case 1 alone: diffusion through a heterogeneous medium

It is worth seeing in detail what the whole framework collapses to when the only active response is D_T^c, because that case is the standard model of diffusion through a spatially varying medium, and the collapse explains a feature of equation (20) that is otherwise easy to misread.

The reduction. Switch off every sensitivity except D_T^c. From table 2 the chemotactic mobility is then $\kappa = -D_T^c$, and it is convenient to give a name to the full spatially varying diffusivity,

$$\mathcal{D}(\boldsymbol{r}) = \underbrace{D_{T0} + D_T^c\,\delta c(\boldsymbol{r})}_{\text{translational noise, responsive}} + \underbrace{\frac{v_0^2}{2\gamma_0}}_{\text{swimming, constant}}. \tag{21}$$

Only the first group varies in space: the second is built from v_0 and γ_0, which are background constants in this case, so self-propulsion merely lifts the whole landscape by a fixed amount. With this definition, equation (19) becomes, to the order retained throughout,

$$\boldsymbol{J}_\rho = -\boldsymbol{\nabla}\big[\mathcal{D}\rho\big], \qquad \text{so that} \qquad \partial_t\rho = \boldsymbol{\nabla}^2\big[\mathcal{D}\rho\big]. \tag{22}$$

The check is one line. Expanding the gradient,

$$-\boldsymbol{\nabla}(\mathcal{D}\rho) = -\mathcal{D}\boldsymbol{\nabla}\rho - D_T^c\rho\boldsymbol{\nabla}c,$$

and the first term equals $-D_{\text{eff}}\boldsymbol{\nabla}\rho$ up to the residual $-D_T^c\delta c\,\boldsymbol{\nabla}\rho$, which is $\mathcal{O}(\varepsilon^2)$ because $\boldsymbol{\nabla}\rho$ is itself $\mathcal{O}(\varepsilon)$ by section 5.2. The second term is exactly $\kappa\rho\boldsymbol{\nabla}c$ with $\kappa = -D_T^c$. So the two expressions agree at the retained order.

If in addition the swimmers do not self-propel, $v_0 = 0$, the identification is exact at the microscopic level too, not merely at the retained order. The full process equation (6) then loses its propulsion term, its tumbles are irrelevant because the heading no longer influences the position, and what is left is

$$\mathrm{d}\boldsymbol{r}_t = \sqrt{2\mathcal{D}(\boldsymbol{r}_t)}\,\mathrm{d}\boldsymbol{W}_t, \qquad \mathcal{D} = D_T,$$

a pure Itô diffusion with multiplicative noise and no drift. equation (22) is its Fokker–Planck equation.

Why the drift was never a drift. The instructive part of equation (22) is that the two pieces of equation (19) — one diffusive, one apparently advective — have merged into a single gradient. The term $-D_T^c\rho\boldsymbol{\nabla}c$ was not an independent transport mechanism. Nothing pushes the swimmers anywhere; they simply take larger random steps in some places than in others, and

under the Itô convention that alone produces a current. Writing the flux as a diffusive part plus a drift, as the closure naturally does, splits one physical effect into two bookkeeping terms.

The consequence for the steady state is the familiar one. Setting $\boldsymbol{J}_\rho = \mathbf{0}$ in equation (22) gives $\mathcal{D}\rho = \text{constant}$, hence

$$\rho_{\text{ss}}(\boldsymbol{r}) \propto \frac{1}{\mathcal{D}(\boldsymbol{r})}.$$

Swimmers accumulate where they move sluggishly, not because anything attracts them there, but because they leave slowly.

The boundary condition. Applying $\boldsymbol{n} \cdot \boldsymbol{J}_\rho = 0$ to equation (22) gives

$$\partial_n \big[\mathcal{D}\rho\big]\Big|_{\partial\Omega} = 0, \tag{23}$$

a homogeneous Neumann condition on the product $\mathcal{D}\rho$ rather than on ρ alone. This is the structural signature of the Itô flux, and it is the practical point of the whole subsection: what is flat at an impermeable wall is $\mathcal{D}\rho$, while the density itself acquires a compensating gradient wherever the medium is heterogeneous. A code that imposes $\partial_n \rho = 0$ here will not conserve the population.

Finally, the two descriptions agree where they should. Expanding equation (23) at the wall gives $\mathcal{D}\,\partial_n\rho + \rho\,\partial_n\mathcal{D} = 0$, and since $\partial_n\mathcal{D} = D_T^c \partial_n c$ from equation (21), this is

$$\partial_n \rho = -\frac{D_T^c}{\mathcal{D}}\,\rho\,\partial_n c,$$

which at leading order is equation (20) with $\kappa/D_{\text{eff}} = -D_T^c/D_{\text{eff}}$, as it must be.